AI IN STRUCTURAL HEALTH MONITORING FOR PREDICTIVE MAINTENANCE & SAFETY ASSURANCE

EXECUTIVE PROGRESS SUMMARY: PRIOR MDP, CURRENT IDP, DUAL-PLATFORM DEPLOYMENT & PROCUREMENT PLAN

BACSE291 -- Innovative Design Project | Fall 2026 | Vellore Institute of Technology (VIT), Vellore

Guide: Dr. Senthil Kumar N (Emp ID: 15377, SCOPE) | **Aerial (15377IDP01):** Priyam S., Ayushman K. | **Ground (15377IDP02):** Nisargkumar J. (Lead), Aryan M., Mudit G.

1. PREVIOUS MDP PROJECT: PLY FORGE POINT CLOUD PROCESSOR

• **Architecture & Independence:** Client-side WebGL 1.0 3D point cloud processor in HTML5/ES6 requiring zero external libraries (no Three.js, no Node.js backend).
• **Core Capabilities:** Binary/ASCII PLY parser (`DataView`/`ArrayBuffer`), built-in 18,000-pt Stanford Bunny demo, circle ROI filtering, 2.5D Delaunay/grid meshing, and PLY export.
• **Voxel Downsampling:** 3D spatial hash map downsampling point clouds by 60%--90% while preserving structural boundaries.
• **Direct IDP Role:** Directly embedded into Ground Station Tab 4 (Streamlit WebGL Viewer) providing zero-latency 3D facade rendering without external desktop software.

2. CURRENT IDP WORK DEVELOPED & VERIFIED

• **Aerial Sensing (15377IDP01):** Calibrated Benewake TF-Luna 850nm dToF (<5g, 0.35W, 100 Hz) on ESP32 FreeRTOS. UART sync to 0x59 0x59 with checksum verification and median filter (N=5).
• **Empirical Metrology Fit:** Universal regression model: $y = 1.0228x + 3.20$ cm (R^2=1.0000, RMSE=±0.02 cm across 39.0cm and 61.4cm physical benchmarks).
• **Ground Analytics (15377IDP02):** Rolling baseline tracking calculating geometric residuals (Δd); flags spalls (Δd ≥ +3.5cm) and bulges (Δd ≤ -3.5cm).
• **3D Digital Twin:** Reconstructed 9,969-point room (.PLY) isolating Almari bulge (+65cm, Orange) and Gate recess (-60cm, Green). 5-tab Streamlit dashboard active.

3. PLATFORM SPECIFICATION: DRONE VS. SMART 4WD CAR

• **Platform Duality:** (1) *Autonomous F450 Drone:* Elevated facade & bridge inspection; (2) *Smart 4WD Ground Car:* Indoor corridors, eliminating prop-wash suction & hand jitter.
• **If Drone Used:** Requires F450 airframe, 4x motors/ESCs, 3S LiPo, Pixhawk autopilot (+INR 17,350) -> Drone Total = **INR 26,124.00**.
• **If Car Used:** Requires 4WD chassis, 1:48 gear motors, dual optical encoders, L298N driver, mast (+INR 3,830) -> Car Total = **INR 12,604.00** (MVP: INR 9,964).
• **Shared Hardware:** YDLIDAR X2 (INR 7,189) + Tilt Sensor (INR 245) + ESP32 (INR 380) + Regulators/Harness (INR 960) = **INR 8,774.00**.

4. FUTURE ROADMAP & SYSTEM INTEGRATION

• **Phase 1--2 (Weeks 1--3):** Procure Tier 1 hardware (LiDAR, tilt sensor, 4WD car kit); verify UART framing and 5V rail regulation.
• **Phase 3--4 (Weeks 4--6):** Assemble F450 airframe & propulsion; implement silicone vibration-damping deck (>18 dB attenuation); benchmark hover thrust.
• **Phase 5--6 (Weeks 7--10):** Laboratory wall scanning with Smart 4WD car (encoder odometry); tethered facade flights; blind defect detection.
• **Phase 7 (Weeks 11--14):** End-to-end integration, Streamlit PLY Forge 3D meshing, automated compliance PDF reports, and IEEE submission.
• **Target Benchmarks:** Static distance RMSE ≤ ±1.5cm; Tilt residual ≤ 0.5cm; Defect sensitivity ≥ 95%; Car odometry drift ≤ 1.0cm / 5m.

Hardware Package	Specific Inclusions & Role	Subtotal (INR)	When Required
1. Shared Core Sensing	YDLIDAR X2 360° LiDAR (INR 7,189), DFRobot Tilt (INR 245), ESP32 MCU (INR 380), Buck/Harness (INR 960)	**INR 8,774.00**	**Required for BOTH Drone & Car**
2. Drone Platform Only	F450 Frame, 4x Motors/ESCs (INR 5,200), 3S 4200mAh LiPo & B6 Charger (INR 4,430), Pixhawk Autopilot (INR 5,950), Reserve	**INR 17,350.00**	**Required ONLY if Drone is Used**
3. Smart 4WD Car Only	4WD Chassis + 4x Gear Motors (INR 1,050), Dual Optical Encoders (INR 320), L298N (INR 380), Mast (INR 650), 12V Li-ion (INR 1,150)	**INR 3,830.00**	**Required ONLY if Car is Used**
TOTAL INTEGRATED DUAL-PLATFORM	**Complete All-Environment System: Shared Core + Drone Platform + Smart 4WD Car Platform**	**INR 28,874.00**	**Car-Only MVP: INR 9,964.00**